

BEVISONNEERS

THE MERCEDES-BENZ FELLOWSHIP

Planet-positive Project Portfolio
BioViL

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What is your business model?

Farm-to-Business-Ecosystem Model

*“Our business model creates a closed-loop ecosystem on each farm by seamlessly integrating a bio-digester into the farmer’s existing **agricultural and business** workflows. By capturing the **business** aspect of the farmer, we unlock a range of energy applications beyond **mere cooking fuel** - expanding **revenue** opportunities while substantially reducing environmental impact. This holistic approach drives profitability and fosters **long-term** sustainability for both farmers and communities.”*

Revenue Channels:

- **Mixed Funding (Grants + Equity)**
We use the grants to reduce initial costs for farmers with **20% of additional own contribution** in **each project** to ensure long term financial returns.
- **Revenue-Sharing with farmers**
Our cooperative model allows farmers to take **full-ownership** in the project, while we are responsible for the energy sales and share profits, creating a sustainable revenue loop for both parties.
- **Carbon-Credits Monetization**
By capturing and reducing methane emissions, we earn verifiable carbon credits that we sell to organizations looking to meet **ESG (Environmental, Social, Governance)** goals.
- **Surplus Selling**
We will supply the surplus biogas to nearby homes and businesses at prices significantly lower than the existing fuel, i.e. LPG, eliminating costly bottling or transport. Existing LPG stoves can be easily retrofitted for biogas, making adoption both affordable and convenient.

How will people discover your project?

- **Local Agricultural Cooperatives**
We'll partner with co-ops to organize educational workshops and demo days, using their networks and trust to reach potential adopters. This will also help us engage the youth of the region to be involved with the projects and potentially take lead in such initiatives.
- **Online Platforms and Social Media**
We'll post short videos on how our **Farm-to-Business-Ecosystem** based bio-digesters work, sharing testimonials from satisfied farmers. Along with this we also plan to launch a website to also broaden our message to further potential partners.

- **Word-of-Mouth**

Early adopter farmers will be encouraged to share their experiences with fellow farmers, neighbours, creating organic buzz and trust around our solution and approach

How do you know you're successful?

Although there are numerous ways to know if the project is going as planned, we have defined certain KPIs to track the same:

- **KPI-1: Energy production & Utilization**

Measuring the biogas output and monitoring how much is locally used, and how much is exported to neighbouring consumers, indicating **real impact** on **energy self-sufficiency rate**.

- **KPI-2: Levelized Cost of Energy(LCOE)**

The levelized cost of energy will be analysed and compared to the initial cost of the energy usage for both the farmer and the neighbouring homes and businesses.

- **KPI-3: Revenue Growth & Cost Savings**

We look at both our own revenue increases and the cost savings for the end-users (farm, homes, etc) - if both are rising, we know our model delivers shared value.

- **KPI-4: Economic Viability & ROI**

Tracking the payback period for the farmer, plus savings for the farmer, and revenue from surplus energy sold to neighbours.

- **KPI-5: Reduction in Environmental Impact**

We monitor weekly, monthly, and annual GHG (greenhouse gas) reductions through documented methane capture and utilization, aiming for clear, measurable cut in emissions

- **KPI-6: Adoption Rate**

We track how many community members, and farmers are interested in the idea after the workshops, learning sessions, etc and are interested in investing or having the system implemented at their farm.

- **KPI-7: Retention & Satisfaction**

We run periodic surveys to gauge user satisfaction; a high retention rate for the different events, programs, or services indicates long term viability.



BIOViL



Weaving all the findings
together

In summary

Throughout this journey, my project evolved significantly. Initially, my idea was **to build biogas systems for rural households to reduce firewood use**. Since then, **BioViL** has transformed into a **structured three-step model: Educate** communities about bioenergy, **Aware** them of its benefits, and **Implement** solutions tailored to their unique needs

My biggest insights or unexpected lessons included:

- Traditional biodigesters often fail not because of technology flaws, but due to **lack of motivation** or **operational mismanagement** by users.
- Many existing programs install biodigesters without adapting to a **farm's infrastructure** or ensuring **direct financial returns** for farmers, leading to abandonment.

The most significant challenges I overcame were:

Gaining trust with farmers and communities. After months of dialogue and conversations, we secured an **agreement** with a farmer to use his land and organic waste to install BioViL's first biodigester. This system will serve both his household and neighboring homes.

Moving forward, the next steps for my project will be:

1. **Pilot Execution:** Install and operate the first biodigester by late 2025, demonstrating its viability.
2. **Community Momentum:** Use this success to inspire broader adoption of clean energy practices across the region.

To publicly share my project, I created a LinkedIn post (tagging @bevisioneers), highlighting:

- **My journey:** Key milestones like securing the farmer partnership and refining our three-step model.
- **Why it matters:** Turning organic waste into energy empowers farmers to lead change while improving health and reducing deforestation.
- **Invitation:** Encouraging entrepreneurs, NGOs, and organizations to collaborate on scaling BioViL's impact.